IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Falcon 9 booster landing analysis | IBM Data Science Capstone



Outline

Executive summary and research question

Methods: collection, wrangling, SQL, EDA, mapping and dashboards

Results from exploratory analysis and classification

Conclusions, limitations and project files

Executive Summary

I analyzed 90 historical Falcon 9 launches using IBM course data (2010–2020).

The overall recorded booster landing success rate is 66.7% (60/90).

I cleaned the records, explored locations, payloads and orbits, and used four classification models.

Logistic regression, SVM and KNN each reached 83.3% accuracy on the same test set.

Introduction

SpaceX develops reusable rocket boosters. This project looks at the first-stage landing outcome rather than mission success.

Question: Can historical launch information help explain and predict whether a booster lands successfully?

I compare flight order, payload mass, launch site, orbit and booster features.

Methodology

Section 1

Methodology

Section 1



Methodology at a Glance

Data: IBM SpaceX API snapshot and Wikipedia scraping exercise.

Cleaning: missing values, outcome labels and one-hot categorical encoding.

Analysis: pandas, SQL, seaborn/matplotlib, Folium and Plotly Dash.

Prediction: train/test split, scaling, grid search with 10-fold CV, test accuracy.

Data Collection

SpaceX API archive → JSON records → pandas DataFrame.

Wikipedia HTML → BeautifulSoup → launch tables.

The API server returned HTTP 525 during execution; the archived IBM JSON/CSV was used to continue.

I kept the data-source distinction in the report.

SpaceX API Data Collection

1. Load historical launch JSON from IBM course mirror.
2. Normalize the nested JSON into a DataFrame.
3. Identify Falcon 9 launches and launch attributes.
4. Keep 90 launches for the later exercises.

Notebook and executed outputs: see the linked GitHub project.

https://github.com/DanielEspinosaLuengo/IBM-Data-Science-Professional-Certificate/tree/main/10-Applied-Data-Science-Capstone

Notebook/source: jupyter-labs-spacex-data-collection-api-v2.ipynb

Wikipedia Web Scraping

1. Download the Falcon 9 launch-history HTML.
2. Parse launch tables with BeautifulSoup.
3. Extract dates, booster versions, payloads, launch sites and outcomes.
4. Handle missing hyperlinks by extracting cell text.

The scraping exercise uses a separate historical table from the API snapshot.

https://github.com/DanielEspinosaLuengo/IBM-Data-Science-Professional-Certificate/tree/main/10-Applied-Data-Science-Capstone

Notebook/source: jupyter-labs-webscraping.ipynb

Data Wrangling

The final modeling table contains 90 observations.

Map landing outcomes to binary Class: 1 = successful landing; 0 = other outcome.

Fill missing payload mass where required; inspect missing landing-pad values.

One-hot encode orbit, launch site, landing pad and booster serial.

The resulting feature table has 80 columns.

https://github.com/DanielEspinosaLuengo/IBM-Data-Science-Professional-Certificate/tree/main/10-Applied-Data-Science-Capstone

Notebook/source: labs-jupyter-spacex-Data wrangling-v2.ipynb

EDA with Visualization

Scatterplots compare flight number and payload mass across launch sites and orbits.

A bar chart summarizes success rate by orbit.

A line chart shows the yearly landing-success pattern.

I use color to distinguish successful and unsuccessful booster landings.

https://github.com/DanielEspinosaLuengo/IBM-Data-Science-Professional-Certificate/tree/main/10-Applied-Data-Science-Capstone

Notebook/source: jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

Queries used SELECT DISTINCT, WHERE, LIKE, GROUP BY, COUNT, SUM, AVG, MIN/MAX and date conditions.

Questions included unique launch sites, NASA payload mass, early landing dates and outcome counts.

The exercise database is a separate launch-history table and is not identical to the 90-row API sample.

https://github.com/DanielEspinosaLuengo/IBM-Data-Science-Professional-Certificate/tree/main/10-Applied-Data-Science-Capstone

Notebook/source: jupyter-labs-eda-sql-coursera_sqllite.ipynb

Interactive Map with Folium

Site markers show launch-site positions and landing outcomes.

Color-coded markers make success and failure easier to compare.

Circles, lines and proximity checks help explore geographic context.

The HTML map is saved as a separate project artifact.

https://github.com/DanielEspinosaLuengo/IBM-Data-Science-Professional-Certificate/tree/main/10-Applied-Data-Science-Capstone

Notebook/source: lab-jupyter-launch-site-location-v2.ipynb

Plotly Dash Dashboard

Dropdown: All Sites or a selected launch site.

Pie chart: share of successful landings by site or outcome split at one site.

Payload slider filters the mass range shown in the scatterplot.

Point colors show booster version categories.

The screenshots included later were captured from the running app.

https://github.com/DanielEspinosaLuengo/IBM-Data-Science-Professional-Certificate/tree/main/10-Applied-Data-Science-Capstone

Notebook/source: spacex_dash_app.py

Predictive Analysis

80 input features and binary outcome target.

Standardize predictors, split into train and test sets.

Tune logistic regression, SVM, KNN and decision tree using GridSearchCV.

Compare all models on the same held-out test records.

Accuracy reflects this historical course sample, not future rocket performance.

https://github.com/DanielEspinosaLuengo/IBM-Data-Science-Professional-Certificate/tree/main/10-Applied-Data-Science-Capstone

Notebook/source: SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Results — Overview

90 Falcon 9 launches, 60 recorded successful booster landings.

Success varies across launch sites and orbit categories.

Yearly success becomes more frequent in later years, with fluctuations.

Three classification models tied at 83.3% test accuracy.

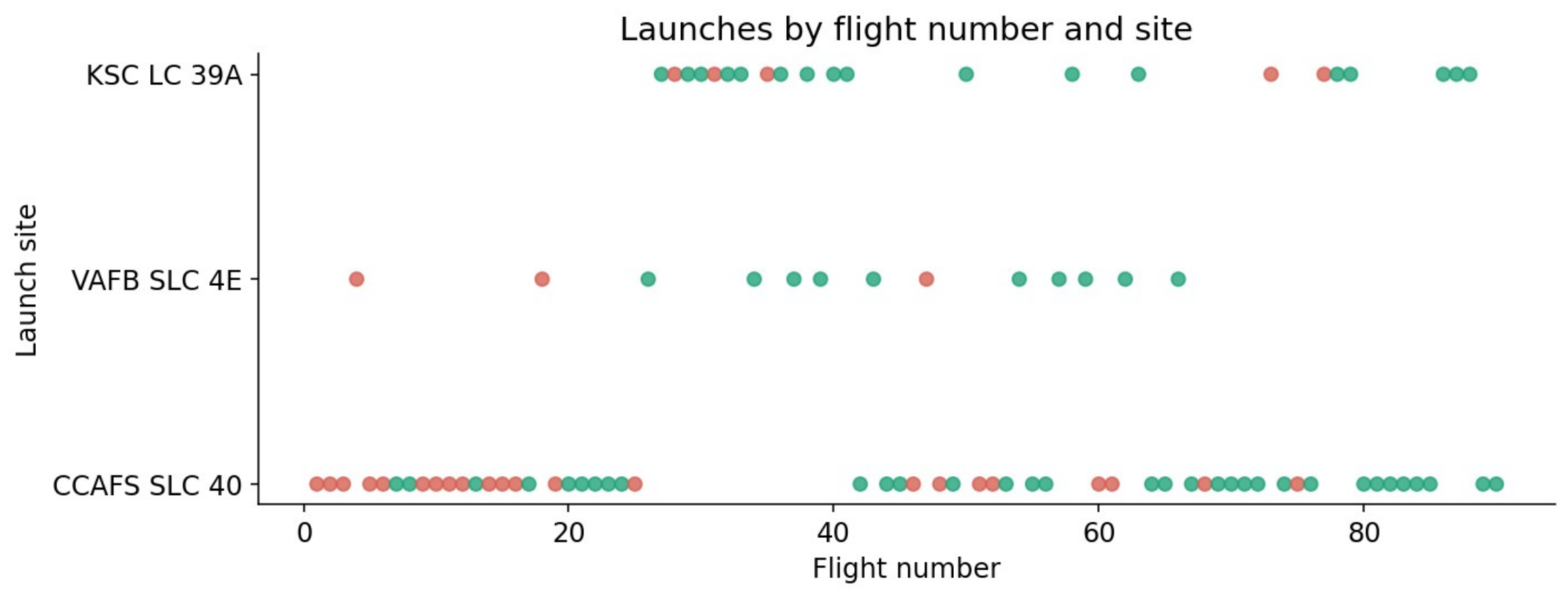
Section 1

Exploratory Analysis

Section 2

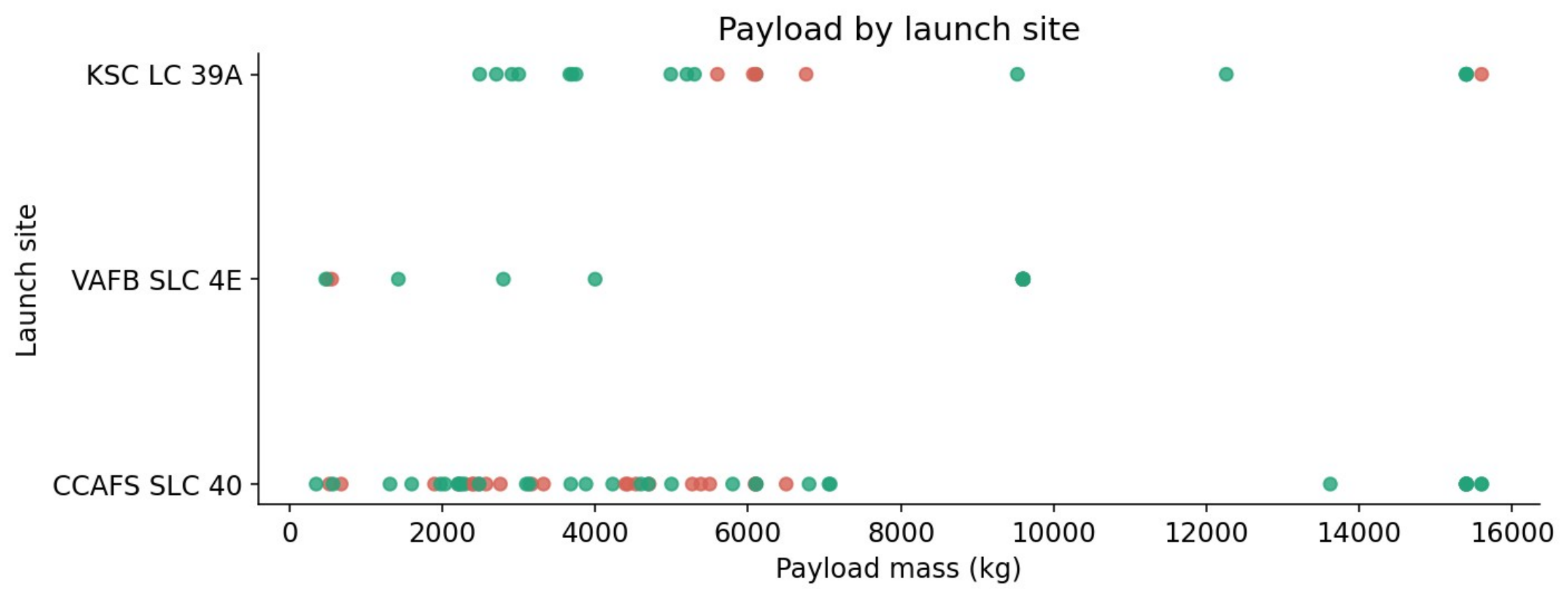
Insights drawn from EDA

Flight Number vs. Launch Site



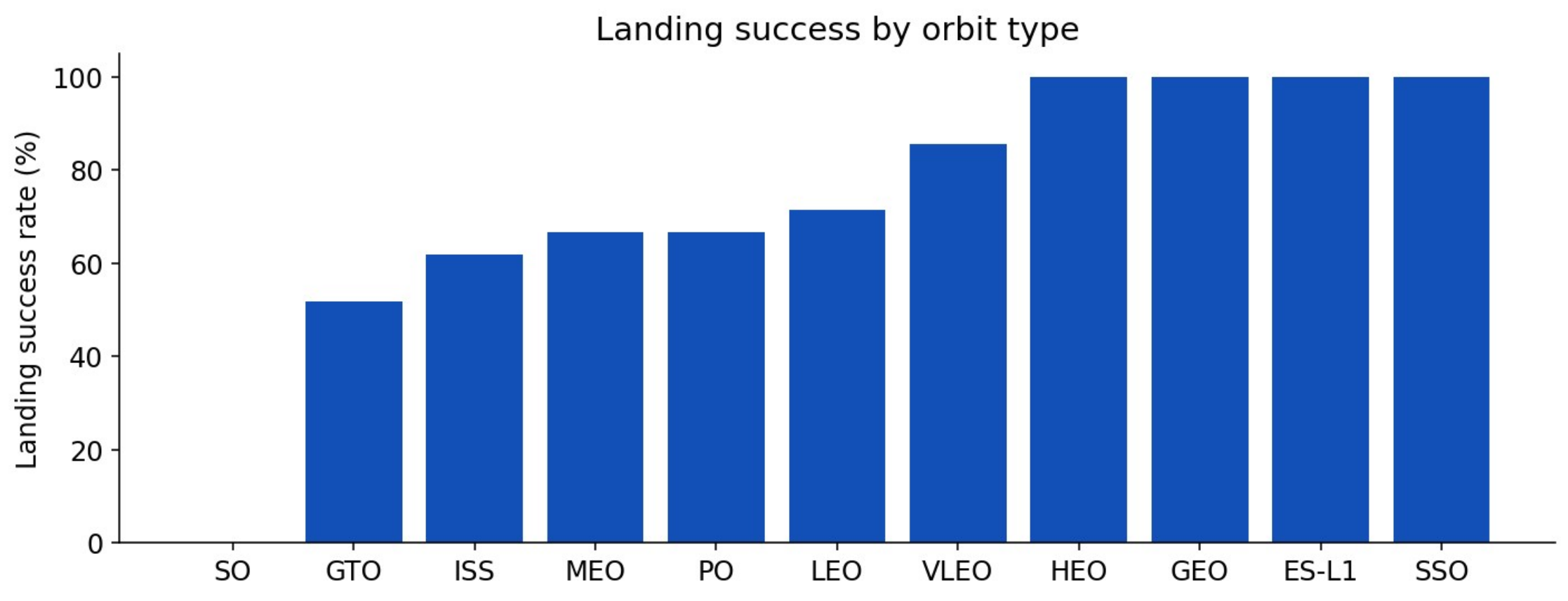
Green = successful landing; red = other outcomes.

Payload vs. Launch Site



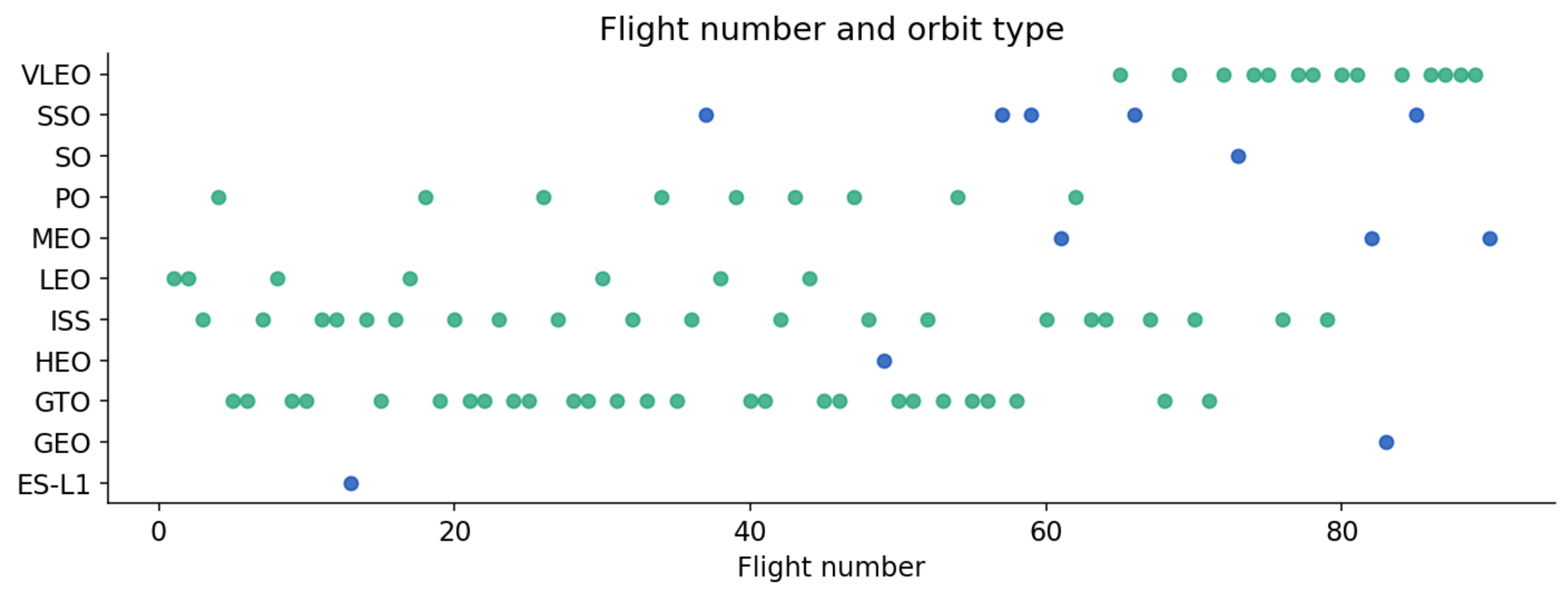
Payload distributions vary across the three sites.

Landing Success by Orbit



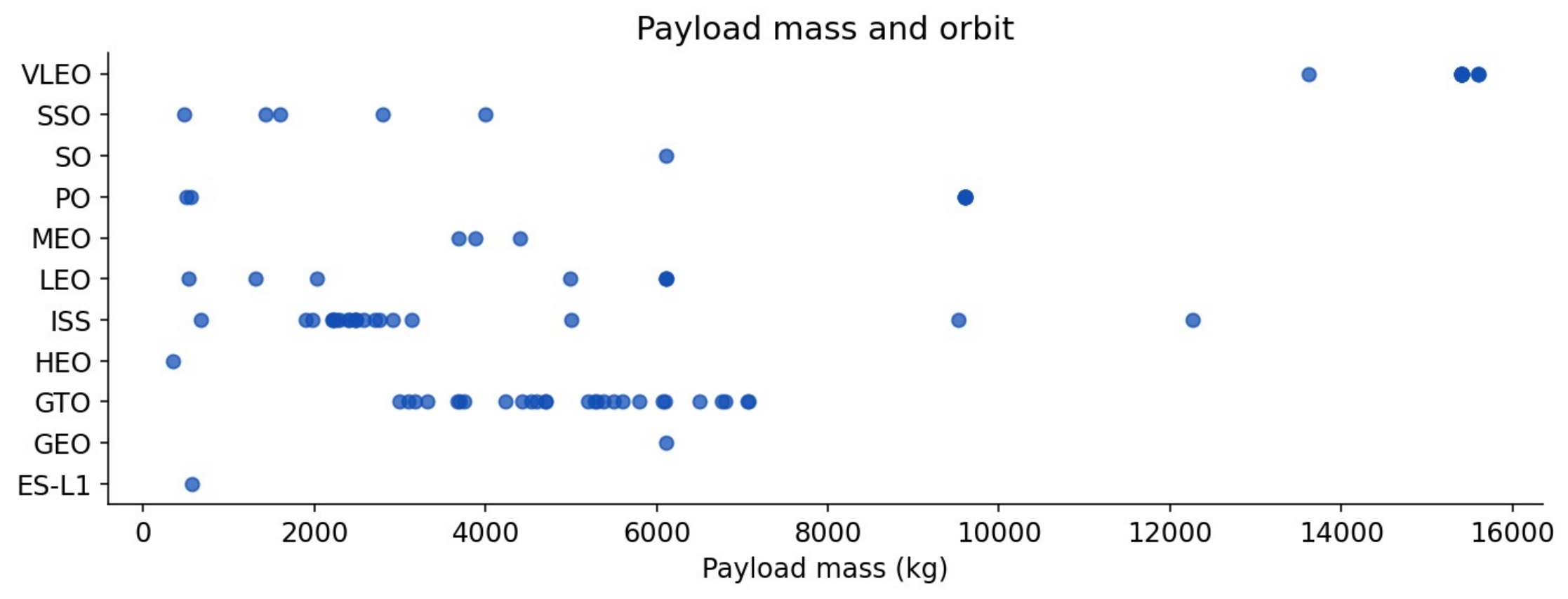
Small orbit categories should be interpreted carefully.

Flight Number vs. Orbit



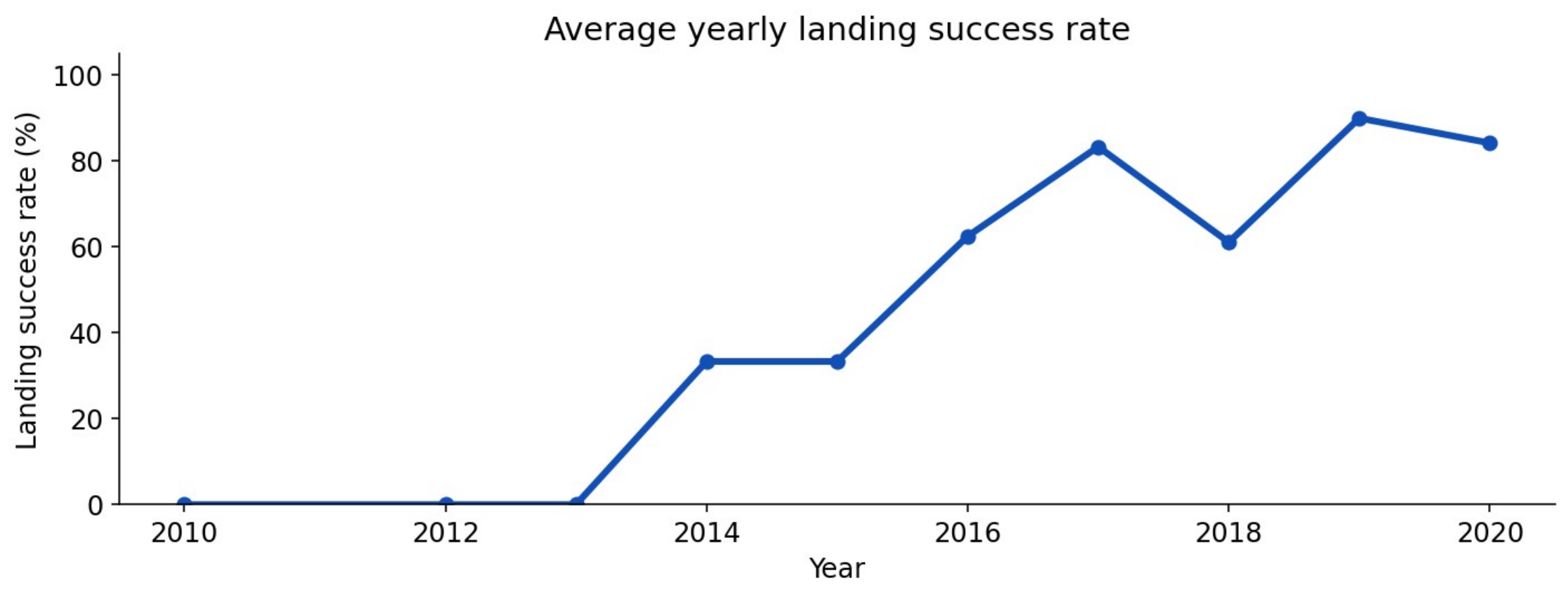
Launches are grouped by orbit type and flight order.

Payload vs. Orbit



Payload masses differ by orbit category.

Yearly Success Trend



Rates tend to improve but are not monotonic.

Unique Launch Sites

SQL: SELECT DISTINCT Launch_Site FROM SPACEXTBL;

The historical SQL table contains CCAFS LC-40, CCAFS SLC-40, KSC LC-39A and VAFB SLC-4E naming variants.

The cleaned API sample consolidates the main launch sites into three groups.

Query result — distinct launch sites

CCAFS LC-40 · CCAFS SLC-40 · KSC LC-39A · VAFB SLC-4E

The two CCAFS names are kept separate because they are distinct strings in this historical table.

Launch Sites Beginning with CCA

SQL: SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE "CCA%" LIMIT 5;

This filters launches from Cape Canaveral using a prefix match.

The LIMIT clause displays five example records rather than the full set.

Query result — first five matching rows

2010-06-04 | CCAFS LC-40 | F9 v1.0 B0003

2010-12-08 | CCAFS LC-40 | F9 v1.0 B0004

2012-05-22 | CCAFS LC-40 | F9 v1.0 B0005

2012-10-08 | CCAFS LC-40 | F9 v1.0 B0006

2013-03-01 | CCAFS LC-40 | F9 v1.0 B0007

NASA Payload Mass

SQL: SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer LIKE "%NASA%";

The sum answers how much payload mass NASA-associated missions carried in the SQL table.

Note: the database includes rows not identical to the 90-row modeling sample.

Query result — SUM(payload mass)

107,010 kg of payload mass across rows whose Customer field contains “NASA”.

Source: saved SQLite table SPACEXTBL (historical IBM course dataset).

Average Mass — F9 v1.1

SQL: SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE "F9 v1.1%";

This estimates typical payload mass for the early Falcon 9 v1.1 booster version.

Query result — AVG(payload mass)

2,534.67 kg average payload for booster descriptions beginning with “F9 v1.1”.

First Successful Ground Landing

SQL filters Landing_Outcome = "Success (ground pad)" and returns MIN(Date).

First successful Falcon 9 booster landing on solid ground: 2015-12-22 (UTC date in this course table).

Query result — MIN(Date)

2015-12-22 — earliest “Success (ground pad)” recorded in the SQL table.

Successful Drone-Ship Landings

SQL filters Landing_Outcome = "Success (drone ship)" and payload mass BETWEEN 4000 AND 6000 kg.

This demonstrates filtering by both numeric payload interval and landing outcome.

Query result — four qualifying launches

JCSAT-14 (4,696 kg) · JCSAT-16 (4,600 kg)

SES-10 (5,300 kg) · SES-11 / EchoStar 105 (5,200 kg)

Mission Success and Failure Counts

SQL uses TRIM(Mission_Outcome), GROUP BY and COUNT(*).

Mission outcome refers to carrying out the flight mission; it is not the same variable as first-stage booster landing success.

The course SQL data contains extra outcome variants.

Query result — normalized mission outcomes

Success: 99 · Failure (in flight): 1 · Success (payload status unclear): 1

A trailing space in one “Success ” entry was removed for grouping.

Maximum Payload Mass

SQL selects MAX(PAYLOAD_MASS__KG_) and the booster versions with this value.

This highlights a record payload launch in the SQL dataset.

Maximum payload in the cleaned 90-launch API dataset: 15,600 kg.

Query result — maximum payload

15,600 kg (MAX(PAYLOAD_MASS__KG_)). This is the highest recorded payload in the SQL exercise table.

2015 Drone-Ship Landing Failures

SQL combines year filtering on Date with Landing_Outcome = "Failure (drone ship)".

The output provides booster version and launch site for the selected unsuccessful landing attempts.

Query result — 2015 drone-ship failures

F9 v1.1 B1012 — CCAFS LC-40

F9 v1.1 B1015 — CCAFS LC-40

Landing Outcomes over Time

SQL groups outcomes between 2010-06-04 and 2017-03-20 and orders counts from most to least frequent.

The output distinguishes ocean-controlled, ground-pad and drone-ship outcomes.

Query result — outcomes in selected date range

No attempt 10 · Success (drone ship) 5 · Failure (drone ship) 5

Success (ground pad) 3 · Controlled (ocean) 3 · Uncontrolled (ocean) 2

Failure (parachute) 2 · Precluded (drone ship) 1

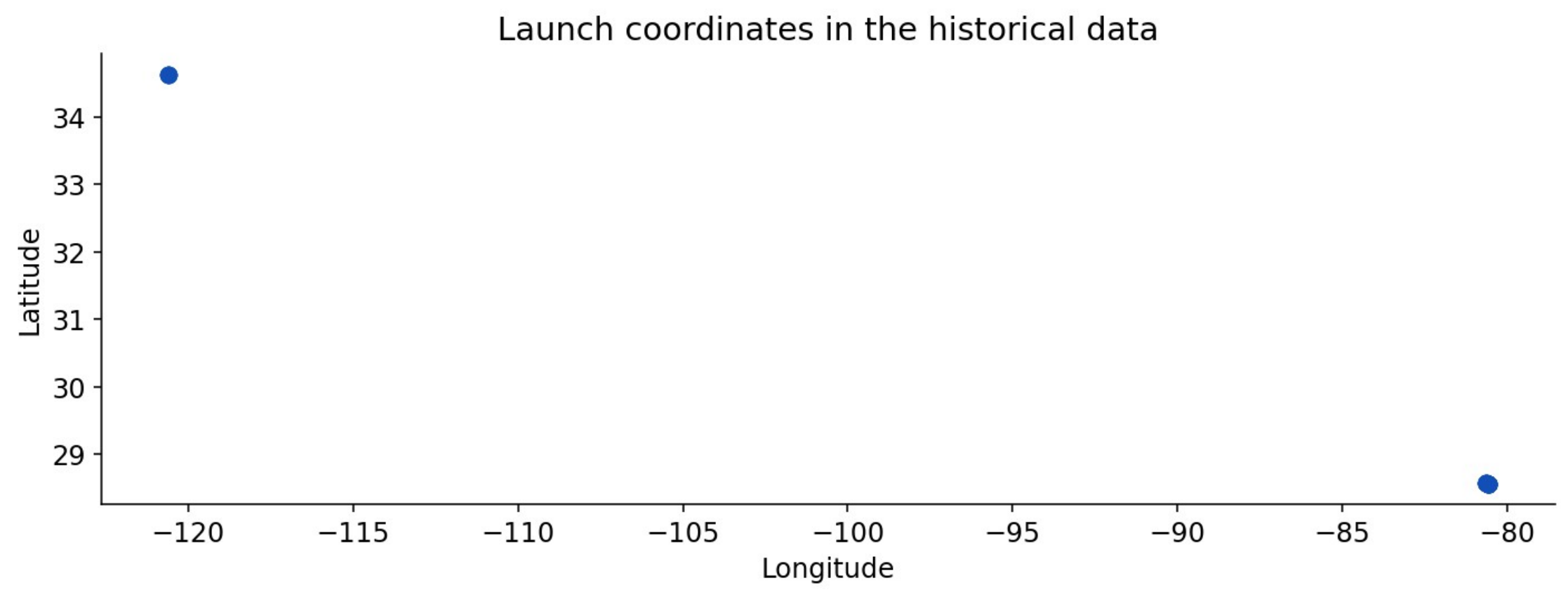
Section 4

Interactive Folium Maps

Section 3

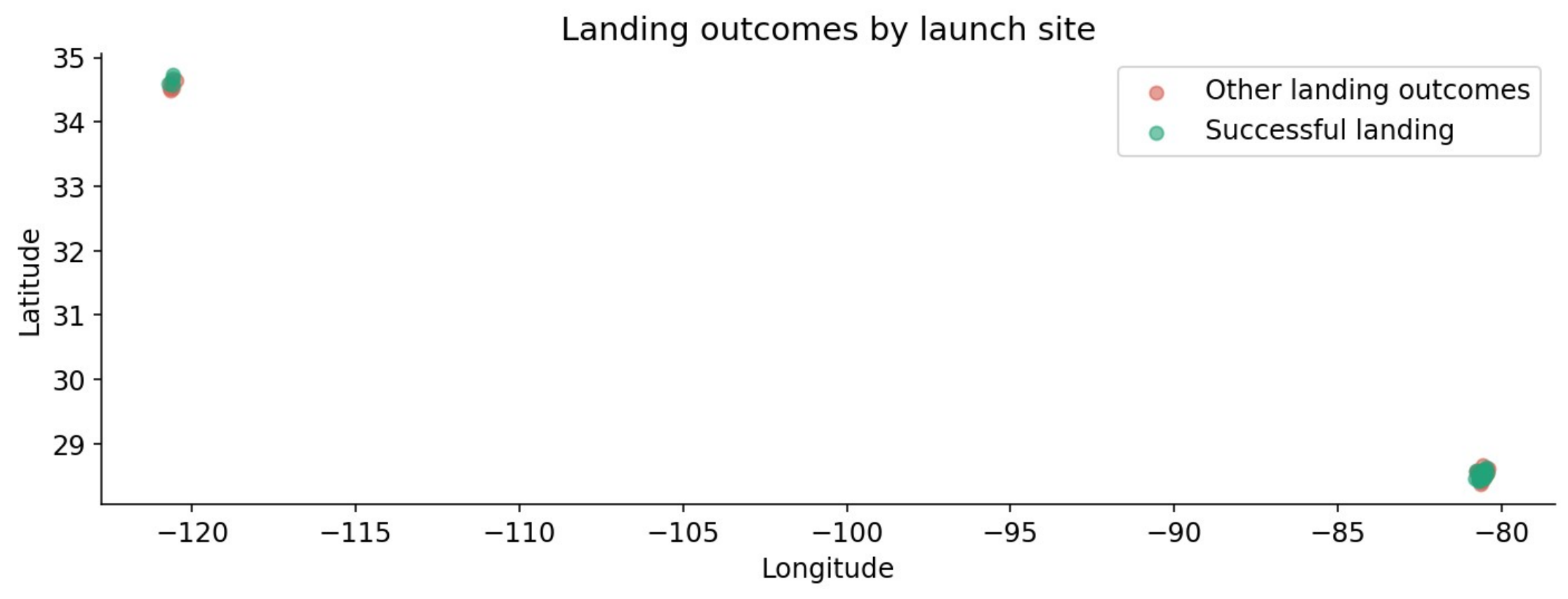
Launch Sites Proximities Analysis

Launch Site Map



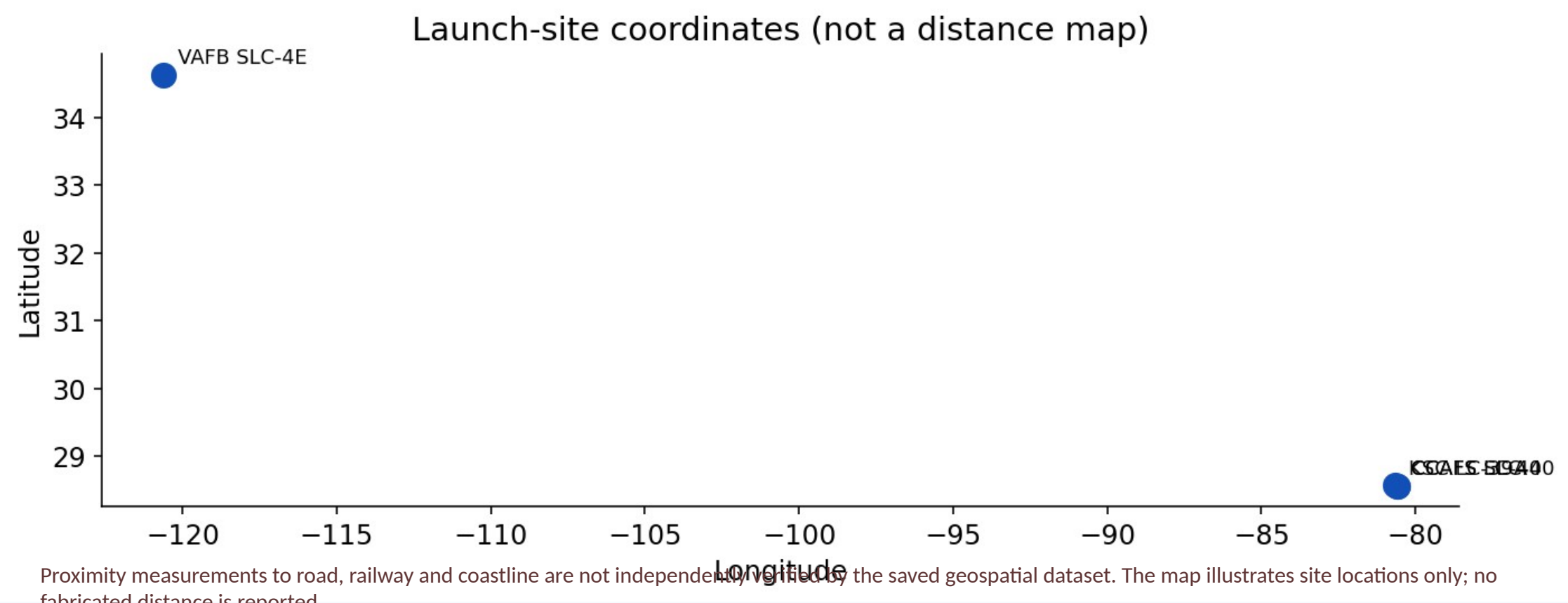
Historical coordinates; site markers overlap where many flights share one site.

Launch Site Outcomes



Colors indicate landing class, with slight jitter to separate identical coordinates.

Launch-Site Geography



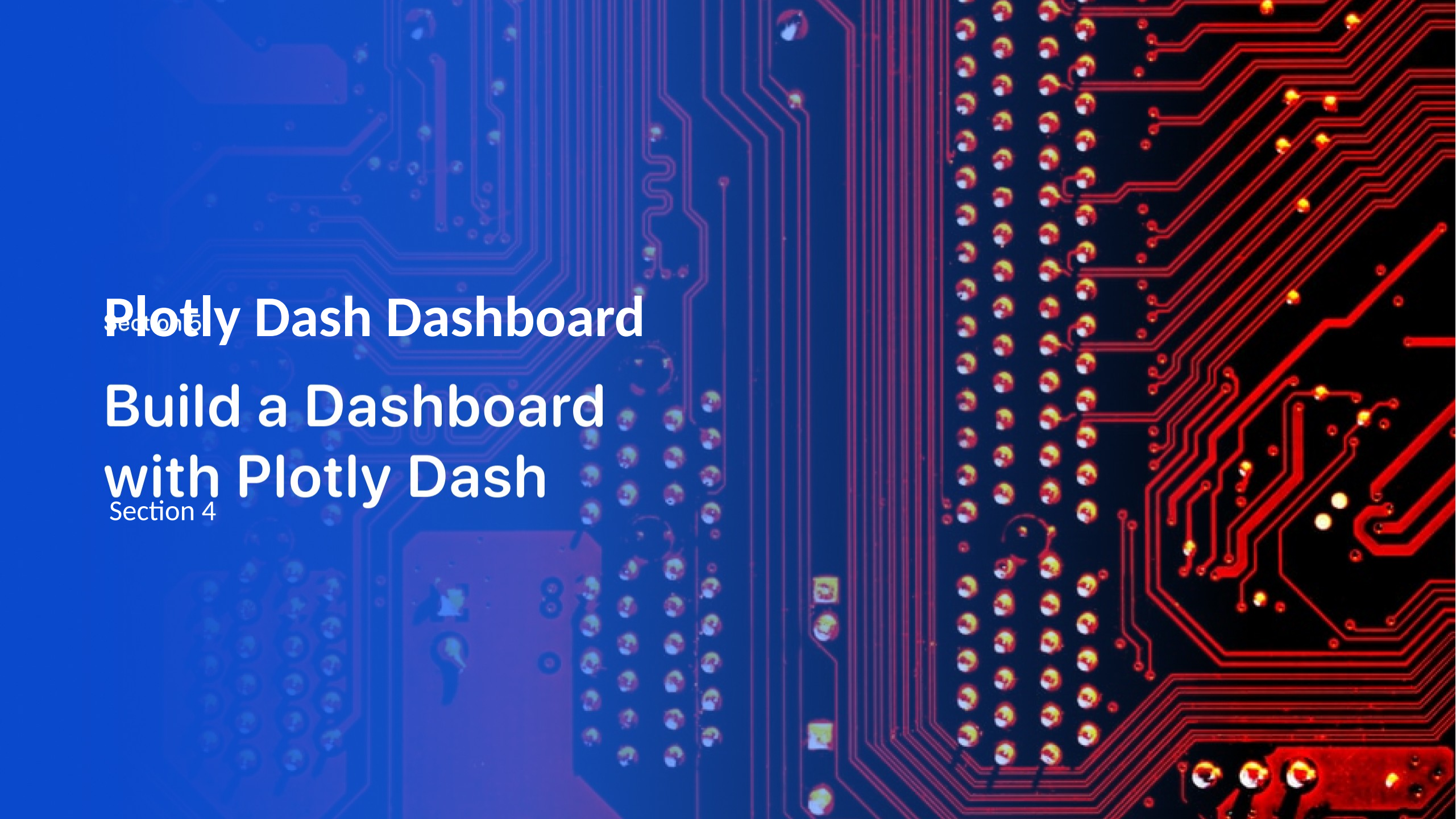
Proximity measurements to road, railway and coastline are not independently verified by the saved geospatial dataset. The map illustrates site locations only; no fabricated distance is reported.

Section 5

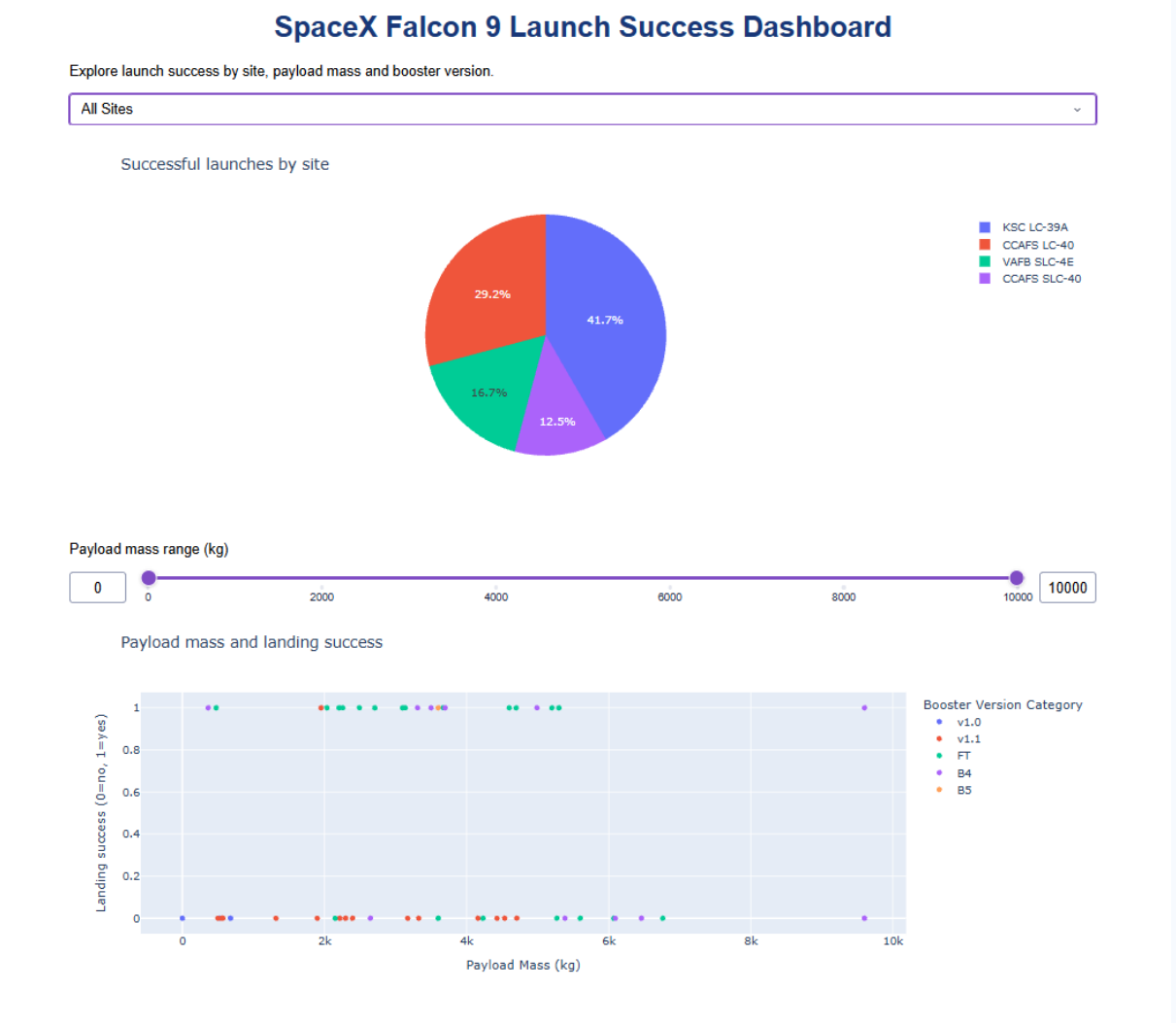
Plotly Dash Dashboard

Section 4

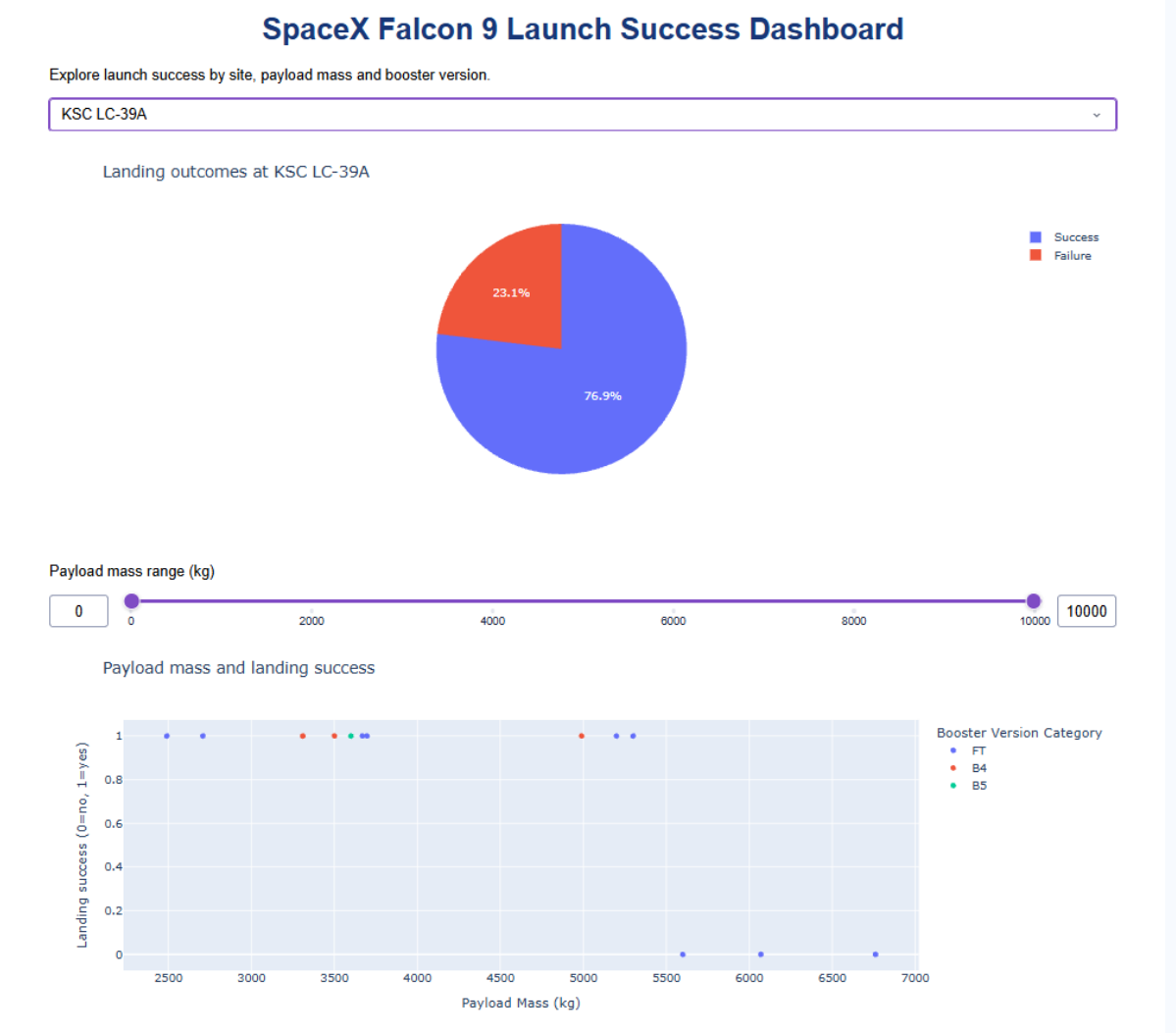
Build a Dashboard with Plotly Dash



Dashboard — All Launch Sites



Dashboard — KSC LC-39A



Dashboard Interactions

Changing the launch-site dropdown updates the pie chart and scatterplot.

The payload slider filters the points shown by mass.

The two screenshots compare a general view with one selected site.

Section 6

Predictive Analysis

Predictive Analysis

Predictive Analysis (Classification)

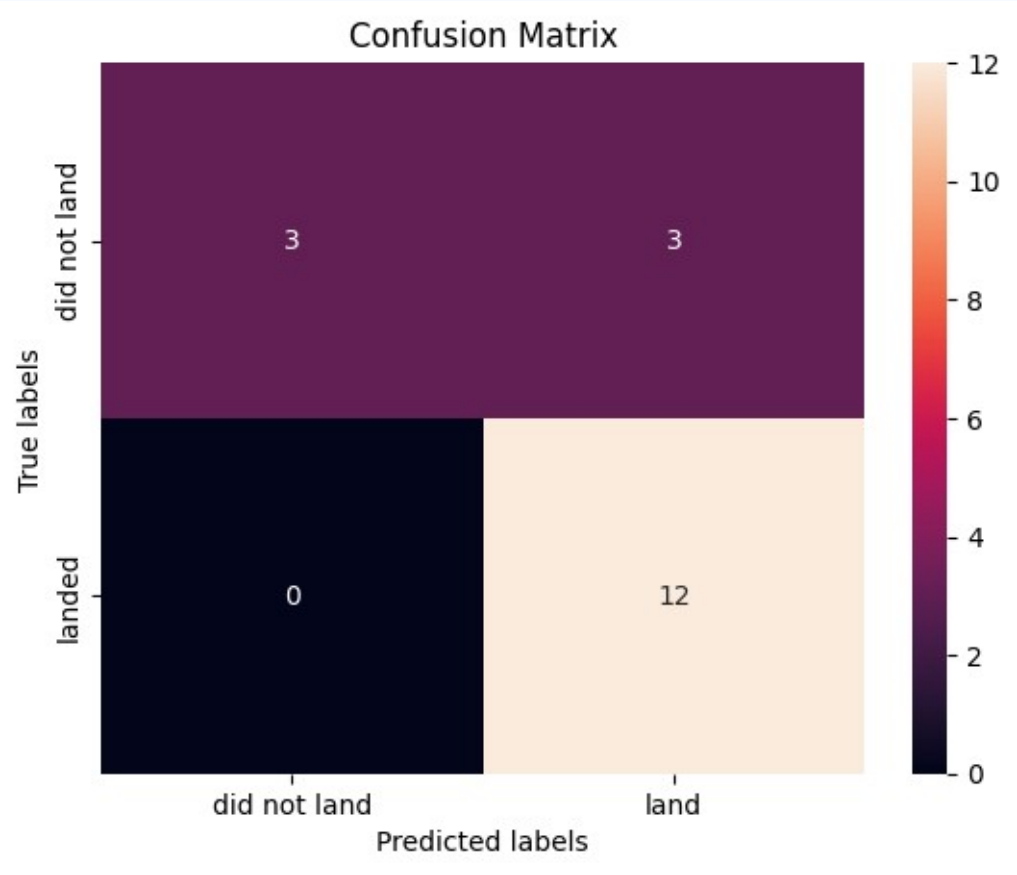
Section 5



Classification Model Comparison

Three models tie on the same test split; decision tree performs worse.

Confusion Matrix — Interpretation



Logistic regression confusion matrix from the executed notebook. 12 true positives; 3 false positives. Test accuracy: 83.3%.

Conclusions

Landing success became more common in the later years of this historical dataset.

The landing rate differs by orbit and site; smaller categories can give unstable percentages.

Logistic regression, SVM and KNN tied at 83.3% accuracy on the test split.

These patterns are associations, not proof that payload or launch site causes success.

More recent launches and external validation would improve confidence.

Appendix — Reproducibility

Source notebooks: API, web scraping, data wrangling, SQL, visual EDA, Folium and ML.

Saved files: dataset_part_1.csv, dataset_part_2.csv, dataset_part_3.csv, SQLite database and Folium HTML.

Plotly Dash Python application and two saved browser screenshots.

All course artifacts are linked from the GitHub folder.

Dataset note: SQL exercise table contains 101 rows; modeling dataset contains 90 rows. These are different course snapshots and should not be directly equated.

References and Limitations

GitHub: github.com/DanielEspinosaLuengo/IBM-Data-Science-Professional-Certificate/tree/main/10-Applied-Data-Science-Capstone

This is a historical IBM course dataset, not a live SpaceX flight database.

The API endpoint was unavailable (HTTP 525); the IBM archived snapshot was used.

Differences in site naming and source populations were not silently merged.

Model comparisons are descriptive for a small held-out sample.